

Replicating *Deep Learning in Asset Pricing*

Chen, Pelger & Zhu, *Management Science* 70(2), 2024 — MFE 230ZA final project

Data, method, challenges and extensions of a from-scratch PyTorch replication of the GAN stochastic discount factor

1. Data

What the paper uses. Monthly CRSP excess returns for all US stocks with a complete set of 46 firm characteristics (past returns, investment, profitability, intangibles, value, trading frictions), January 1967 – December 2016. Characteristics are rank-transformed cross-sectionally every month into $[-0.5, 0.5]$. The macro information set is 178 time series: 124 FRED-MD variables (McCracken–Ng transformations), the 46 monthly cross-sectional medians of the characteristics, and 8 Welch–Goyal predictors. Split: train 1967–86 (240 months), validation 1987–91 (60), test 1992–2016 (300).

What we used. CRSP/Compustat are behind WRDS, so we used the *processed panel the authors released* (Google Drive folder linked from Pelger’s Stanford page): `Char_{train,valid,test}.npz` = $(T \times N \times 47)$ arrays with the excess return in column 0 and -99.99 marking stock-months with missing characteristics, and `macro_*.npz` = $(T \times 178)$. The panel is *unbalanced*: 430–2177 stocks per month in training (336k stock-months), 1933–2826 in the test sample (750k). Macro series are standardized with training-sample moments only.

2. Method

Economics. No arbitrage implies an SDF $M_{t+1} = 1 - \sum_i \omega_{t,i} R_{t+1,i}^e$ whose weights are a function of the information set, and the conditional moment restriction $\mathbb{E}[M_{t+1} R_{t+1,i}^e g(I_t, I_{t,i})] = 0$ for *every* conditioning function g . The paper turns “for every g ” into a two-player game (a generative adversarial method of moments):

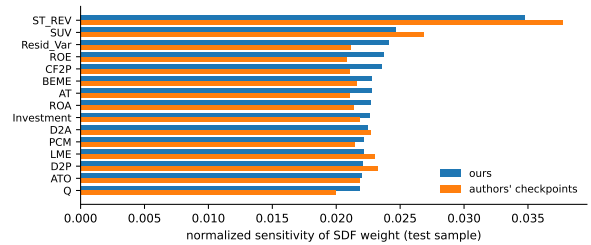
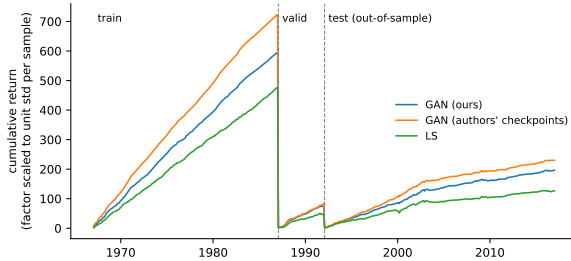
$$\min_{\omega} \max_g \frac{1}{N} \sum_{i=1}^N w_i \left\| \frac{1}{T_i} \sum_{t \in \mathcal{T}_i} M_{t+1} R_{t+1,i}^e g(I_t, I_{t,i}) \right\|^2, \quad w_i = T_i / \max_j T_j.$$

Networks (exactly the authors’ optimal configuration). *SDF network*: an LSTM compresses the 178 macro series into 4 hidden states h_t ; a feed-forward net $[46 \text{ chars}, h_t] \rightarrow 64 \rightarrow 64 \rightarrow 1$ (ReLU) outputs $\omega_{t,i}$. *Adversary*: a second LSTM (32 states) plus one tanh layer outputs 8 test-asset instruments $g_{t,i}$. Dropout 0.05, Adam 10^{-3} , full-batch gradient steps (4 per epoch). *Schedule*: (1) 256 epochs on the unconditional moments ($g \equiv 1$); (2) 64 adversary steps that raise the moment violation on the best-validation-loss SDF; (3) 1024 epochs of the SDF against the fixed adversarial instruments. The paper reports that further GAN rounds do not help, so this is one full round. *Selection and ensembling*: the checkpoint with the highest validation Sharpe ratio is kept; nine random seeds are trained and their ω ’s averaged, then L^1 -normalized each month. *Metrics*: Sharpe ratio of the SDF factor $F_{t+1} = \sum_i \omega_{t,i} R_{t+1,i}^e$; explained variation (EV) and cross-sectional R^2 from a monthly projection of returns on loadings $\beta_{t,i} \propto \mathbb{E}_t[R_{t+1,i}^e F_{t+1}]$, which the authors estimate with a second feed-forward net ($46 \rightarrow 32 \rightarrow 16 \rightarrow 8 \rightarrow 1$ predicting $R_{t+1,i}^e F_{t+1}$). *Benchmarks*: the linear special case LS ($\omega = \theta^\top I_{t,i}$, closed form on 92 long/short characteristic-managed portfolios) and a ridge version of the paper’s elastic net (EN).

3. Results

Model	Sharpe ratio (monthly)			EV		XS- R^2	
	train	valid	test	train	test	train	test
GAN – ours, retrained from scratch (9-seed ensemble)	2.48	1.29	0.66	0.17	0.06	0.14	0.22
GAN – ours, TF-loop-faithful variant (3-seed ensemble)	2.58	1.34	0.73				
GAN – authors’ checkpoints run through our code	3.02	1.39	0.77	0.16	0.07	0.13	0.21
GAN – paper, Table I	2.68	1.43	0.75	0.20	0.08	0.12	0.23
UNC ($g \equiv 1$, no adversary) – ours / paper	1.52	0.70	0.45 / 0.53				
Ridge (EN) – ours / paper	1.77	1.24	0.49 / 0.50		0.03 / 0.04		0.15 / 0.19
LS – ours / paper	1.98	0.79	0.42 / 0.42		0.03 / 0.03		0.15 / 0.14

Single seeds are noisy (test Sharpe 0.15–0.63 across our 9 default trials); the ensemble is what reproduces the paper, and the variant that mirrors the TF training loop (Section 4) closes most of the remaining gap. Annualized ($\times \sqrt{12}$), the out-of-sample GAN Sharpe is 2.3 for our default retrain and 2.7 for the authors’ checkpoints (paper: 2.6), against ≈ 1.5 for the linear model.



Left: cumulative SDF-factor return, unit standard deviation within each sample. Right: sensitivity of the SDF weight to each characteristic (test sample), our ensemble vs the authors' checkpoints.

4. Challenges and how we dealt with them

- **The official code does not run.** It is pinned to TensorFlow 1.12 / Python 3.6 (`tf.placeholder`, `tf.contrib`), which cannot be installed on Apple Silicon or any current Python. We re-implemented the model in ≈ 250 lines of PyTorch from the paper's equations, using the TF source only to pin down undocumented details (loss weights $T_i / \max T_j$, L^1 weight normalization, the 3-phase schedule, best-validation-Sharpe selection).
- **Proving the re-implementation is faithful.** We read the authors' nine TF-1 checkpoints with TF2's checkpoint reader and mapped them into our model. This needs the LSTM gate order (TF: i, j, f, o ; PyTorch: i, f, g, o) and TF's implicit $+1$ forget-gate bias. The ported ensemble gives test/validation Sharpe 0.77/1.39 vs 0.75/1.43 in Table I. The authors also published their SDF-factor time series (`Data.xlsx`); it reproduces Table I to three decimals, and our ported factor has correlation 0.87 (test) / 0.92 (train) with it – so the port is right and the released checkpoints are simply not the exact Table I ensemble. Our own retrain has correlation 0.83 (test) with the published factor.
- **Unbalanced panel and LSTM state continuity.** Stocks enter and leave, so a $(T \times N)$ mask drives the SDF sum, the per-stock moment means $(1/T_i)$ and the loss weights; and the validation/test LSTM states must start from the state at the end of the previous sample, not from zero – resetting it silently destroys the out-of-sample numbers.
- **Seed noise and model selection.** Individual seeds range from 0.15 to 0.63 test Sharpe (validation 0.53–1.00; the authors' own nine checkpoints: 0.45–0.62 test, 0.82–1.22 validation); training Sharpe keeps rising while validation Sharpe plateaus (`fig_training.pdf`). Best-validation-Sharpe checkpoints plus an ensemble of *weights* is what makes the result stable.
- **Undocumented training-loop details.** The TF loop re-samples the adversary's instruments with dropout at every step, selects checkpoints on the *unnormalized* factor's validation Sharpe, never resets that tracker between phases, and starts phase 3 with fresh Adam moments; none of this is in the paper. Our default run fixes g after the adversary step and selects on the normalized factor. Switching all four back (the "TF-loop-faithful" row) lifts per-seed validation Sharpe to 0.98–1.16 (test 0.49–0.69) and a 3-seed ensemble to 0.73 out of sample – these loop details, not the architecture, account for most of the gap to Table I.
- **Compute.** Full-batch training over 336k stock-months for 1 280 epochs $\times$ 4 steps plus a validation pass every epoch (the test sample is never touched during training): ≈ 35 min per seed on one CPU core, ≈ 12 min on an Apple-silicon GPU (MPS); seeds were queued three at a time on the GPU.
- **EV and XS- R^2 are not free.** They need the second-stage β network described only in the code (target $R_{t+1,i}^e F_{t+1} \times 50$); the cross-sectional R^2 in Table I is the T_i -weighted variant.

5. Extensions

- **Transaction costs (done).** With the paper's definition (Table A.VII) the test-sample turnover of our SDF portfolio is 0.40 for the long and 0.97 for the short leg (paper: 0.40 / 1.02); the short leg is rebuilt almost entirely every month. Charging 10 / 25 / 50 bp per unit of dollar turnover (0.94 per month) cuts the test Sharpe from 0.66 to 0.57 / 0.45 / 0.24. A cost-aware version would add a turnover penalty to the objective, which turns the static weight map $\omega(I_t, I_{t,i})$ into a genuine control problem with the previous position as a state.
- **Stability over time (done).** Test Sharpe is 0.83 in 1992–2004 vs 0.47 in 2005–16 and 0.71 excluding 2008–09; worst month -4.2%. The decay is the familiar post-publication erosion of characteristic premia.
- **Ablations run for free.** The unconditional model (no adversary) gives test Sharpe 0.45 and the linear model 0.42, isolating what the adversarial test assets and the nonlinearity buy.
- **Reinforcement-learning reformulation.** $\omega(\cdot)$ is a policy mapping state $(h_t, I_{t,i})$ to actions; maximizing the SDF Sharpe by policy gradient (actor-critic, with the adversary as a critic that finds mispriced test assets) links the paper to the course and handles costs and constraints natively.
- **Extend the sample and the state model.** Re-run on 2017–2025 data to measure out-of-sample decay; swap the LSTM for a linear-Gaussian state-space model with a Kalman filter to test whether 4 nonlinear states beat 4 linear factors.

Code in `replication/` (`gan_sdf.py` is the model); data from `mpelger.people.stanford.edu/data-and-code`; paper arXiv:1904.00745.